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RESEARCH

Employment scams: a crime script and scoping review of prevailing methods, trends and future directions

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Abstract

Employment scams, where victims are defrauded by applying to fake or deceptive job advertisements, have become a growing concern since Covid-19. Many jurisdictions, especially financial centres such as the United States, Hong Kong and Singapore, have reported year-on-year increases in employment-related deception. Economic pressures from the pandemic, as well as the growth of industrialised scam compounds in Southeast Asia that rely on labour trafficking, have accelerated this growing crime problem. Academic research on this trend, particularly its detection through algorithmic pattern detection, has existed for some time but remains scattered across different disciplines. A holistic structured analysis of existing research, and our current scholarly understanding of this issue and its detection, is therefore necessary to inform future approaches and priorities. This study has two complementary aims: first, it conducts an exploratory PRISMA-ScR-compliant scoping review that analyses 16 research articles relating to the commission and detection of employment scams. Second, the insights gained are used to construct a crime script, which provides a holistic understanding of how employment scams occur and possible points of intervention. Using the findings from both the review and the script, the paper analyses key findings in contemporary employment scam research, key scam typologies, prevailing research limitations, available datasets/methods and useful avenues for future study. The aim of this paper is to provide a unified overview of the current state of the art of both employment scams and current research, while defining the next best steps to support the prevention of this worsening crime trend.

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Introduction

In January 2025, famous Chinese actor Wang Xing flew to Bangkok, having accepted an advertisement for a casting call. He was picked up at the airport and abducted, instead being taken to the border town of Myawaddy in Myanmar (Gan et al. 2025). His abductors were organised criminals operating a scam centre, where thousands of other labour trafficking victims like Wang were forced to commit investment and romance scams on an industrialised scale, under duress (Chen 2025).

Wang Xing was released following a covert cross-border Thai police intervention after three days in captivity, his captors likely concluding that his high profile disappearance was detrimental to their operations (Davidson and Ratcliffe 2025). However, his story raised both the woes and hopes of many other families who had lost loved ones in similar circumstances – they had all accepted supposedly lucrative job offers abroad in the Southeast Asia region, only to be abducted and themselves forced into committing scams (Gan et al. 2025; Lee 2025; Liu 2025).

This example highlights the surge and severity of employment fraud. However, this type of scam is neither new nor limited to facilitating labour trafficking operations. More generally, the move from traditional to digitised job recruitment has created opportunities for efficiencies in the labour market but also new avenues for exploitation by fraudsters. In typical cases, criminals will pose as fake recruiters, often advertising too-good-to-be-true opportunities (Benny 2023; Indeed, 2025). These scam job ads are often placed among legitimate job boards, such as LinkedIn, or disseminated through social media or direct messaging apps – making it difficult for victims to distinguish between what is real and what is fake (Thompson 2024; Whitworth 2024). Scammers may do this to cause financial loss by demanding advanced "processing" fees or deceive victims into being trafficked, similar to Wang Xing. Numerous other illicit end goals associated with employment scams will be discussed further throughout this paper.

A range of global developments have exacerbated this problem. Covid-19, for example, led to increases in unemployment, creating a surge in people seeking jobs during and after the pandemic (Reinicke 2020). The rise in artificial intelligence-enabled solutions, such as text and voice generation, has also made the deployment of scams easier for criminals (Dhaliwal 2024). Recent statistics from major financial centres illustrate the problem: in the United States, scams involving job advertisements increased by 118% in 2023 (Iacurci 2024). The U.S. Federal Trade Commission stated that $367 million had been lost by victims in 2022, up 76% from 2021, with the median loss being approximately $2,000 (Gressin 2023). Other jurisdictions have reported similar increases; in 2023, employment scams in Hong Kong rose by 31% from 2022 (Hong Kong Police Force, 2024a), though remained relatively steady (+4% increase) in 2024 (LegCo Secretariat, 2025).

Research objectives

The overlap between labour trafficking and its recent surge, driven by technological innovation, suggests that employment scams have become a sophisticated and industrialised crime phenomenon serving numerous criminal objectives. This

paper, therefore, seeks to understand the emerging complexities of employment scams by visualising the underlying process as a *crime script*. A crime script is a visual sequencing of a crime process that disaggregates it into a series of individual actions. This allows for: (1) the holistic visualisation of complex processes, typologies and methods in an accessible manner; and (2) the pinpointing of intervention points where criminals appear to be particularly vulnerable (Borrion and Dehghanniri 2016; Cornish 1994; Rowe et al. 2013). Crime scripting has been heavily used across academia and practitioner research for such purposes, for example to script the processes of child sex trafficking (Brayley et al. 2011), theft from oil pipelines (Chainey and Alonso Berbotto 2022), money laundering (Gilmour 2014), drug trafficking (Chiu et al. 2011) and public sector corruption (Rowe et al. 2013).

To devise an employment scam crime script, we use a systematic scoping review approach to extract insights from existing relevant academic research. In addition to a crime script, a scoping review also enables a secondary aim, namely to obtain a qualitative understanding of the current state of the art and what academia has to offer in terms of fighting this growing scam problem going forward. Besides insights into employment scam typologies and trends, we therefore also extract other insights from relevant studies, such as the methodology and datasets they used and key research gaps.

Our study therefore serves as both a visual and analytical synthesis of what is known about employment scams in a manner accessible to practitioners, to visualise the crime process and understand possible intervention points, and to identify core resources, methodologies and future scholarly research directions to address this growing problem. The flowchart in Fig. 1 summarises our research aims and methods.

There is also a future-oriented element to this research. As statistics have shown, the 2020 s have seen a surge in employment fraud. In particular, the rapid evolution of technological and geographical trends motivates the need to remain ahead of the curve, as criminals seek to exploit the latest vulnerabilities in maximising the illicit gains from their scams. This scoping review can be considered an initial desk research component of a "horizon scan", namely a means of understanding potential future scenarios, in this field (GO Science 2024).

The next section introduces and documents the scoping review and crime scripting methodology.

Methodology

About scoping reviews

Scoping reviews are a form of structured review of literature on a topic designed to understand emerging trends in a field, synthesising relevant concepts, categorising relevant evidence for policymakers and identifying core gaps (Arksey and O'Malley 2005). They typically cover a broader scope than systematic reviews (Mellor 2021).

Scoping reviews are common in the nursing and medical sector for reviewing and synthesising various evidence sources for specific treatment practices and proce-

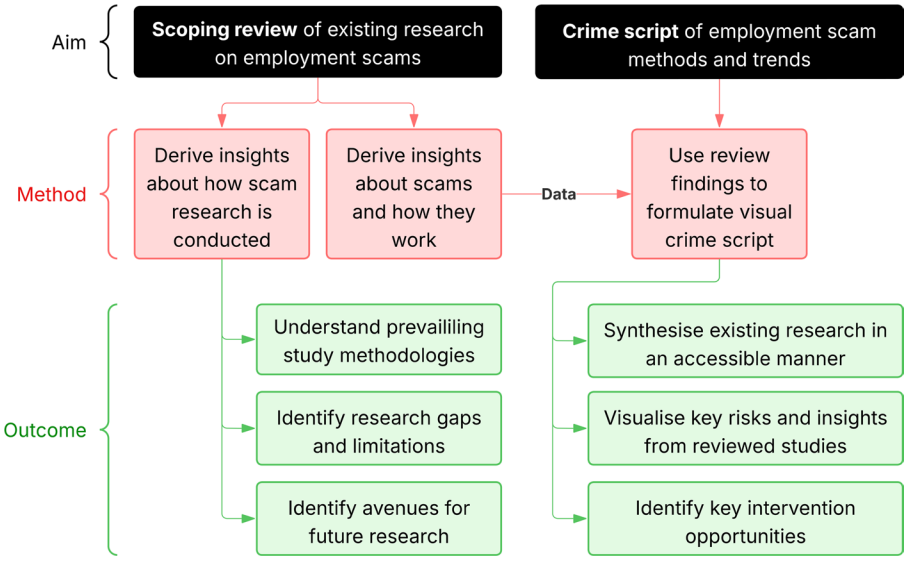


Fig. 1 A flowchart summarising our research aims, methods and intended outcomes

dures. The PRISMA-ScR checklist, with which this review complies, was originally designed to ensure that these medical-oriented reviews were standardised, robust and reproducible (Tricco et al. 2018). In the field of crime science, scoping reviews have been used to derive qualitative data for formulating crime scripts (Akartuna et al. 2024; Chainey and Alonso Berbotto 2022), meaning that our use of this method here is well established. Arksey and O'Malley (2005) devised a methodological framework, which we adopt, for conducting scoping reviews that involved five stages, namely: (1) identifying the research question; (2) identifying relevant studies; (3) study selection; (4) charting the data; and (5) collating, summarising and reporting the results.

This scoping review seeks to answer the following research questions: What is the state of the art of academic research on employment scams, who are the key contributors and what progress has been made in the understanding and prevention of the problem? We also specifically seek to identify insights about how employment scams are conducted to formulate our crime script.

Identifying relevant studies and study selection

The database we have chosen to source relevant studies is ProQuest Central, which is marketed as the "the largest, multidisciplinary, full-text database available in the market today" (ProQuest, n.d.). ProQuest Central contains over 320 million records, including journal articles, working papers, conference papers, reports, books and trade journals, spanning 175 subjects (ProQuest 2014). We therefore consider it a sufficient resource for our scoping review.

Our search aimed to identify any journal articles or conference proceedings relevant to employment scams. We did not limit results by timeframe. Our full search term is as below:

"job scam?" OR "scam job*" OR "employment scam?" OR "scam employment" OR "recruitment scam?" OR "scam recruit?" OR "job fraud" OR "fraud? Job*" OR "recruitment fraud" OR "fraud? recruitment" OR "employment fraud" OR "fraud? employment".

Where "?" indicates wildcard endings, such as "fraudster", "frauds", "fraudulent" for "fraud?" and "" indicates one-letter wildcards, e.g. "jobs" for "job*".*

The search, conducted on 10 April 2024, returned 195 records, which were then subject to hierarchical study selection criteria. The first two criteria related to technical exclusions, namely: (i) studies that were not in English; and (ii) not academic publications. Since the original search already included these filters, only a small number of miscategorised results were removed.

The latter two criteria involved manually reviewing the abstracts and excluding any study that: (iii) did not relate to employment scams or where; (iv) employment scams were not the primary or significant focus of study. This ensured that the remaining publications were relevant in addressing our research aims. The exclusion process was conducted by one primary coder. The flowchart in Fig. 2 visualises the exclusion process.

After completing this process, we subjected a randomised 15% (30) of all the studies to a second coder to test inter-rater reliability (Belur et al. 2018). Among those that the initial coder deemed worthy of inclusion, the second coder agreed with all but one. After a discussion between the first and second coders, the decision was, by mutual consensus, to indeed include this one publication for analysis.

The result was 17 remaining publications, which were then subject to full-text review and data "charting" respectively. Full-text review involved re-applying the latter two exclusion criteria once again to each of the 17 studies based on their entire contents, as opposed to their abstracts only. During this process, one article (which we do not legitimise with a citation) was found to be formed largely of illegible sentences and jargon that bore the hallmarks of automated computerised generation. Indeed, we found that the article had been subsequently retracted by the Journal of Physics Conference Series ('Retraction', 2021) due to findings of "systematic manipulation". We therefore withdrew it from our review. Thus, 16 publications were left for data charting.

All other studies were deemed relevant after this full-text review, meaning that no changes were made.

Charting the data and reporting results

For these remaining 16 studies, we sought to extract and synthesise a range of insights to gauge the overall state of the art on employment scam research. These can be categorised into two types of insights (see Table 1), namely: (1) exploratory and general insights into the nature and conduct of employment scams; and (2) insights arising from the specific research and methods used in each study. The former will be used to devise the crime script, while the latter will be used to better understand the role of academia in formulating solutions to the employment scam problem.

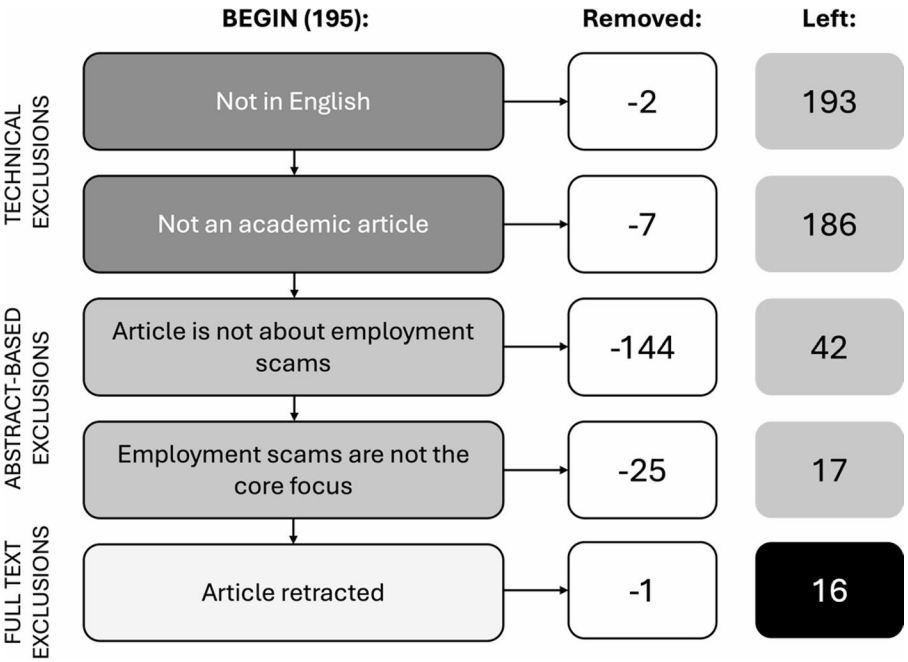


Fig. 2 Flowchart showing the study exclusion process

Table 1 Data charted from reviewed studies

Results	Examples
Exploratory insights	• Employment scam typologies and trends • Key characteristics and methods • Underlying causes and vulnerabilities • Policy implications
Study-specific	• The research aims of each of these papers • Methods • Data sources

The extraction, or "charting", of these insights from the 16 studies was conducted by two data coders for inter-rater reliability purposes. The results section below denotes our syntheses of findings, leading subsequently onto our discussion on how existing studies form the foundation of future research that can assist in devising solutions against this growing crime problem.

Figure 3 shows the number and research aims of each paper included in this review. These aims will be elaborated on in due course. The publication dates indicate a simultaneous rise in publications during the pandemic, where many scams (including employment scams) saw surges across many regions (Johnson and Nikolovska 2024). This indicates that current research has, promisingly, largely reacted according to trends observed throughout this period of growing criminality, and that it remains relevant and not outdated at the time of conducting this review (Q1/Q2-2024).

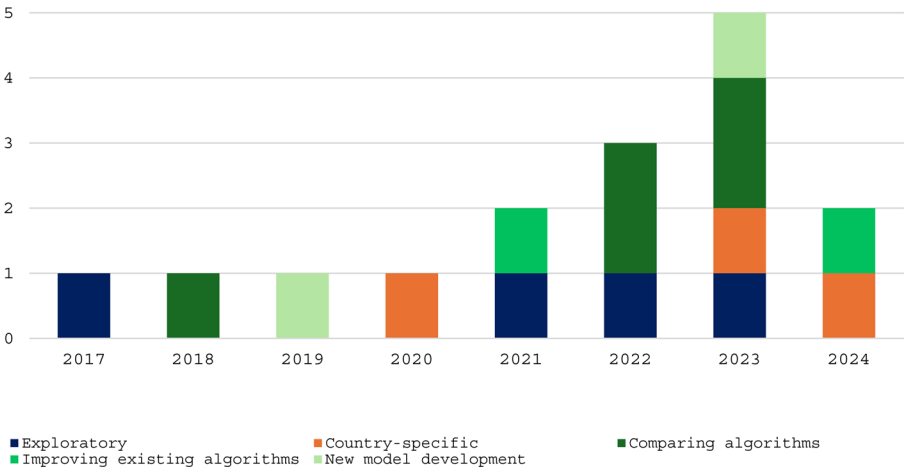


Fig. 3 Number and aims of publications included in the review, by year

Devising the crime script

Following the completion of the review, the exploratory insights from each study were thematically reviewed by one researcher to extract insights into both the motives and steps involved in committing employment scams. Most studies were found to be incomplete in this regard: few explained the full process from start to finish and only made passing or general references to some aspects of the employment scam process. This is a common issue with gathering crime scripting data, and underscores the benefits of reviewing multiple studies so they can complement each other where gaps exist (Akartuna 2023).

Various protocols exist to ensure that crime scripts are constructed objectively (Borrion 2013). One study noted that a single CCTV footage of evidence could spawn 21 unique scripts depending on the scripters' subjective perception of events (Borrion et al. 2017). However, this varies on the detail contained in the evidence, as granularity across reviewed studies may differ based on how specifically they describe the process. Cornish (1994), for example, classified scripts into different types (e.g. metascripts and tracks) depending on how detailed they were.

Cornish (1994) also included distinctive categories for different activities in a script, for example "preparation", "entry" (into crime scene), "doing" (actually committing the crime), "instrumental conditions/initiations" (e.g. selecting and approaching targets), "exit" (leaving crime scene) and so on. However, it is noted that sources of evidence may not have sufficient granularity to categorise in detail. Alonso Berbotto and Chainey (2021) therefore devised a more succinct categorisation, consisting only of "preparation", "pre-activity" (activities that occur immediately before the crime), "activity" (the crime) and "post-activity" (activities that occur after the crime).

Having retrospectively analysed the evidence obtained from our 16 studies, we adopt the categorisation technique of Alonso Berbotto and colleagues, with the researcher thematically extracting employment scam actions mentioned in these studies and categorising them in this manner. Following this process, a second researcher

reviewed the emerging dataset of employment scam actions and resolved any proposed amendments via qualitative discussion.

The emerging actions from the 16 studies were then visualised via *Lucidchart*, an online data visualisation tool for creating flowcharts and mindmaps. The results sections below first set out the exploratory and research-specific insights from these studies, before presenting the completed crime script.

Exploratory insights

This section presents the results of exploratory insights that have been charted from the reviewed evidence, including typologies, trends, characteristics, methods, underlying causes, victim profiles and policy recommendations that publications have identified and associated with employment scams. The results reported here are descriptive and form the basis of the eventual crime script.

Typologies and trends

Characterising a problem is a core step in eventually formulating a solution. We identify that nine of the 16 publications fully disseminate the aims and processes behind what they consider to be employment scams, although it is interesting to note that they often differed in their characterisation. It is also notable that the number of publications that contained no descriptive discussion of the elements of an employment scam constituted a considerable 44% of those that were reviewed.

We identify four broad typologies relating to employment scams, varying in their ultimate aims, in Table 2. Though some publications mentioned two of the typologies, only one publication (Cross and Grant-Smith 2021) mentioned three. None mentioned all four. These typologies and trends are now elaborated, while the created crime script is presented thereafter.

Table 2 The four identified typologies of employment scams[1]

Typology	Explanation	Mentioned by *N*
Advance fee fraud	Asking victim to pay a fee to process their application	7
Data theft	Asking victims for personal details that can then be sold on the illicit data market	4
Labour trafficking	Luring victims under the pretence of starting a job and forcing them into working against their will	3
Money mule scams	Providing work to unwitting victims to assist money laundering by processing illicit financial transactions through their accounts	2

[1]One publication focusing on Nigeria (Ogunmiloro et al. 2024) mentioned two other employment-related frauds that are beyond the scope of this article as they do not necessarily scam victims (but rather, are emblematic of wider corruption that victimises the state as a whole). These were: (1) employment racketeering, namely the process of selling job placements rather than meritocratically selecting the best applicant; and (2) ghost workers, where state funds designed to incentivise employment are pocketed by companies under the guise of paying wages to non-existent workers who exist on paper only

Advance fee fraud

Advance fee frauds (AFF) are so named as they require victims to make an initial payment to continue the supposed application process for a fake job. This was the most mentioned type of employment scam, with 7 of 16 publications discussing them. Although most of these publications discussed AFF in the context of countries such as Nigeria, Indonesia and Cameroon, Wahid (2023) notes that the scam is not limited only to developing nations.

Examples of excuses provided by offenders to legitimise their request for advance fees may include covering the costs of security vetting or admin charges (Srivani et al. 2023). Some fake advertisements may directly demand an upfront fee for the privilege of applying (Wahid 2023). Others may take place over multiple rounds of interactions between the victim and the bogus employer (Ebot 2023).

One publication discussed a more unique means of commission, namely offenders sending the victim fake cheques to cover expenses that contain deliberately overstated amounts (Cross and Grant-Smith 2021). Offenders then request the sum of the 'overpayment' back from the victim.

Numerous publications (Cross and Grant-Smith 2021; Naudé et al. 2023; Ullah and Jamjoom 2023) discussed advance fee fraud as distinct but occasionally intertwined with data theft. Both can often occur synonymously, as the collection of advance fees may also provide scammers with sufficient access to the victim to also harvest their data. The next section considers data theft in more detail.

Data theft

Publications discussing data theft suggest that personal information, such as names, bank details, credit card numbers, social security numbers, ID cards, passport details, phone numbers and emails (Naudé et al. 2023; Ullah and Jamjoom 2023; Vidros et al. 2017). Often, these data are harvested through iterative rounds of interactions, such as online tests, interviews and payment windows, during the "application" process (Ullah and Jamjoom 2023; Vidros et al. 2017).

The purpose of data theft is often to sell stolen ID to other cybercriminals. Dedicated dark web data vendors exist to facilitate the purchases and sales of personal data (Howell and Maimon 2022). Vidros et al. (2017) discuss how access to bank login information can allow criminals to essentially use the victim's account as a "money mule" account, where illicit funds are deposited and withdrawn as a means of laundering their illicit origin. Ullah and Jamjoom (2023) note that, besides facilitating spam or cyberbullying, the application process can also be used to entice victims to download malware or viruses, which can be used to harvest further data from the victim's laptop or mobile devices.

Labour trafficking

Labour trafficking is a criminal phenomenon with a significant pool of dedicated academic research, though specific analyses of how fake recruitment facilitates it has been somewhat less extensive (Volodko et al. 2020). Labour trafficking-based

recruitment scams typically promise victims lucrative jobs overseas, thereby deceiving them into forced and illegal employment that often is completely unrelated to what was initially advertised (Cross and Grant-Smith 2021). As mentioned in the introduction, a growing post-Covid-19 concern has been the rise of dedicated scam compounds in Southeast Asian countries, where workers are duped through employment scams into committing fraud for organised crime groups (Samean et al. 2024).

These trends are not elaborated beyond mere mentions existing studies. Non-academic literature (that was beyond scope of this review) demonstrates clearly the importance of considering this angle for a number of reasons (Bloomberg, 2024; Luong 2025; Reuters 2023). Firstly, the sheer scale of fraud (such as so-called "pig butchering") enabled by dedicated scam compounds in Southeast Asia, staffed substantially through labour trafficking, has been estimated to have reached around US$64 billion annually by some organisations (USIP 2024). Since employment scams are an early-stage enabler of this phenomenon, by virtue of securing labour to execute "pig butchering" and other scams in the first place, they can be considered a critical and early point of intervention that can disrupt this wider criminal ecosystem (Akartuna and Manning 2026).[1]

Secondly, the use of employment scams for labour trafficking, as opposed to direct financial victimisation, suggests that the means of their execution may fundamentally differ in the way they are executed, written, or the platforms on which they appear. For example, some may not be executed online at all – there have been reported cases where trapped workers in cyber scam compounds have lured or sold others to the compounds personally to secure their own freedom, or to make extra money (GITOC, 2025; U.S. Department of State, 2025; Winch et al. 2023). That the execution strategy may be different motivates the need for clearer distinction of employment scam research going forward, demarcated based on the objective of the scams being considered. This is especially relevant for statistical or algorithmic studies: not considering execution differences based on objective (trafficking vs. financial) is likely to manipulate training datasets and therefore reduce the validity of emerging detection algorithms.

Money mules

Money mules refer to individuals – who may be deliberately complicit or unsuspecting – that pass proceeds of crime through their financial accounts as a means of disassociating it from its illicit origin (in other words, money laundering) (CFPB, 2024). Sophisticated crime groups may pass funds through multiple accounts iteratively to make it less traceable. To aid the process, criminals will post job advertisements for "financial transaction managers" or other such roles (Europol, 2020). Mules will typically keep a small portion of the incoming funds as commission.

Money mule positions are often advertised as "financial transaction managers" or other such terms, often with remote working opportunities and a small commission for every transaction that passes through the mule's account. Whether or not this

[1] This underscores the value of considering scoping review results in a crime script format: even where the volume of research is light, crime scripts can reveal opportunities for early interventions and thus, indirectly, priority avenues for further research.

recruitment process can be considered a "scam" is therefore contestable; in terms of paying the mule as advertised, the job advertisements are genuine. However, the fact that the mule is essentially complicit in criminal activity because of their work is naturally kept secret, which can be considered deliberate deception.

Srivani et al. (2023) suggest that students are particularly vulnerable to being targeted to become money mules due to the attractiveness and simplicity of the job. Vidros et al. (2017), meanwhile, consider money mule risk somewhat synonymously with data theft, as the obtaining of bank account information can effectively give scammers access to said accounts to funnel illicit proceeds through (provided that the victim does not notice).

Offender and victim profiles

The reviewed studies do not offer insights into the profiles of the offenders involved with employment scams, besides the organised criminal nature of those engaging in labour trafficking. However, more in-depth insights are offered on the profiles of victims. For example, Ogunmiloro et al. (2024) notes that Nigeria-based victims are often unemployed individuals with a lack of exposure to scam red flags. Cole (2022) identified several characteristics that contribute to exploitation, including a lack of education, low self-control, and an increased willingness to take financial risks. It was noted that individuals from low socioeconomic status (SES) backgrounds tend to be less familiar with cybersecurity protocols, making them less likely to secure their personal data online and more likely to become victims of employment fraud. Wahid (2023) noted this to be the case in Indonesia in particular.

Cross and Grant-Smith (2021) note the lack of public data as a challenge to determining clear offender and victim profiles. However, their analysis suggests an increased likelihood of victimisation by the underemployed (e.g. those on part-time work that seek full-time employment). Furthermore, they suggest that specific attributes, such as LinkedIn users with the #opentowork profile graphic, may make them a target.

Geographic attributes also affect victim profiles. Ebot (2023) argues that the promise of high-paying US company jobs are particularly effective on victims of Indian, Pakistani, Chinese and Filipino nationality, likely due to the significant pay differences observed across these jurisdictions. However, such effects are relative; similar levels of susceptibility have been identified in Romania, Bulgaria, Poland, Slovakia and Lithuania, which between 2010 and 2012 had more victims scammed by fake job posts than any other member state of the European Union (Volodko et al. 2020).

Besides geographic and socioeconomic factors, age and education status also exhibit effects. For example, Srivani et al. (2023) noted that fresh graduates are often more likely to be scammed as they are willing to pay extra money to secure a job, especially a supposedly "low-effort" role as a "financial transaction manager".

Characteristics of employment scams

Several studies went further in depth to explore how employment scams were disseminated or deployed. Naudé et al. (2023) found that short firm profiles were more strongly correlated with identity theft vacancies, while corporate identity

theft positions typically had longer descriptions. This suggests that scammers may use brevity as a tactic to make their fake job postings less conspicuous. Moreover, Cross and Grant-Smith (2021) observed that fake job posts tend to offer excessively large financial compensation in exchange for flexible labor that requires little to no training or expertise. These fake jobs also have twice the rate of remote work compared to real job vacancies, potentially to target a wider pool of vulnerable job seekers.

Regarding the platforms used, scammers typically leverage social media platforms like LinkedIn to publish fictitious advertisements, often pretending to be recruiting firms or headhunters (Srivani et al. 2023). Volodko et al. (2020) analysed specifically the risk indicators of overseas recruitment scams targeting Lithuanian workers that may conceal labour trafficking intent. They found that the most common indicators included the provision of accommodation, provision of relocation assistance, provision of transport, deduction of costs from wages and shared worker accommodation. Certain sectors, such as food production, packaging, hospitality and construction were also identified as high risk.

The tactics used by scammers to commit online recruitment fraud are also documented in some of the reviewed literature. The most frequently mentioned tactic is making these fake ads indistinguishable from real job postings (Srivani et al. 2023; Vidros et al. 2017). Through social learning and adaptation, fraudsters engage in online employment scams by portraying themselves as reputable employers and featuring unreal job opportunities that involve disclosing sensitive information (Cole 2022). Ebot (2023) describes that scammers start with winning the trust of their victims, and then provide further "proof" of legitimacy via gathering information from other channels. Subsequently, they create fake websites mimicking legitimate companies to mislead their targets and present advance payments as necessary expenses (e.g., flights, quarantines, or insurance) that must be paid before a buyer's goods can be shipped.

Online job scammers frequently target recent graduates and students, who are particularly vulnerable to scams due to various psychological factors, including cognitive biases, financial pressures, and limited job market experience. Cognitive biases such as optimism bias may lead individuals to believe they are less likely to fall victim to scams (Owen et al. 2024), while the scarcity effect heightens urgency around job offers, prompting impulsive decision-making (Zhao and Tomm 2018). The sunk cost fallacy further complicates their judgment, as victims often continue investing time and money in scams after an initial commitment (Ridho 2024). For instance, in Indonesia, advance fee fraud scammers typically begin by requesting small amounts, such as registration fees or security deposits, before promising lucrative positions, creating a sunk cost that triggers a distorted psychological response from victims (Wahid 2023). Additionally, the inexperience of recent graduates makes them less capable of identifying warning signs in job offers, while cognitive overload during stressful job searches impairs their judgment (Qudrat-Ullah 2025). Finally, Srivani et al. (2023) suggest that students and recent graduates may be more willing to pay extra fees for job opportunities, further increasing their susceptibility to scams.

Causes and vulnerabilities

We have already discussed, in our introduction, recent factors that may have exacerbated employment scams, such as Covid-19, artificial intelligence and labour trafficking in Southeast Asia. Here, we review whether these causes have also been observed by the reviewed publications, or whether alternative causes have been suggested.

Firstly, general unemployment due to economic constraints, especially among the youth, is a major driving force in many jurisdictions, as desperate job seekers become vulnerable targets (Ogunmiloro et al. 2024; Wahid 2023). Moreover, the methods currently used to detect fraudulent employment are often ineffective and lack scalability (Naudé et al. 2023), thereby providing criminals with increased incentives and motivations to continue their operations. Studies indeed also point to the Covid-19 pandemic as a key exacerbator of the situation, leading to widespread job losses and a sense of desperation on which fraudsters readily capitalise (Cross and Grant-Smith 2021).

Compared with traditional job-seeking platforms, criminals are adept at exploiting user-friendly social media platforms and convenient e-pay methods, allowing criminals to reach a vast audience with expertly crafted job postings in a relatively short time (Ebot 2023; Afzal et al. 2024; Srivani et al. 2023; Volodko et al. 2020). In the meantime, the scarcity of reliable platforms to identify illegitimate job advertisements, the demand for cheap labour in certain regions, and the limited technological capabilities to detect fraud all contribute to the persistence of this problem (Ullah and Jamjoom 2023; Wahid 2023; Wang et al. 2022).

Compounding these challenges is the widespread lack of awareness and knowledge about online recruitment fraud among job seekers, especially those from lower socioeconomic backgrounds (Cole 2022; Wahid 2023; Afzal et al. 2024) – indicating a potential area for greater emphasis for existing anti-scam awareness measures.

Devising the crime script

Using these exploratory insights, we devise the crime script shown in Fig. 4, incorporating our findings relating to actions, typologies, offender/victim profiles, motives and post-activities. Alonso Berbotto and Chainey's (2021) categorisation is used to conceptualise the different actions.

This script shows the four key typologies identified, as well as the range of post-activity crimes that it enables, for example labour trafficking, industrialised scams, cyberbullying, spam, identity theft, dark web sales, money mule account opening and, ultimately, the laundering of both employment scam proceeds and the proceeds of subsequent crimes. It shows a blend of optional and mandatory actions to power the criminal process – offering insights for law enforcement in terms of where intervention points may lie.

We end this section with a review of possible interventions as recommended by the reviewed studies, as they form the basis of this crime script. While doing so, we draw on the crime script to assess whether these recommendations may be effective.

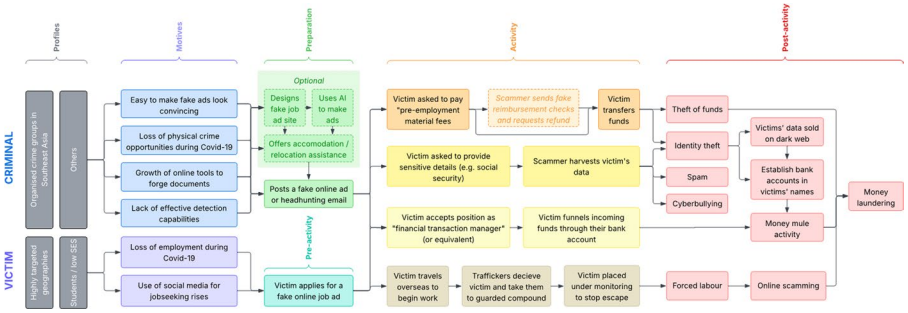


Fig. 4 Crime script of employment scam typologies and methods

Policy Implications, points of intervention and recommendations

The reviewed research offers a comprehensive set of policy recommendations from various perspectives to address employment scams. These can be categorised into three types: technology, government, and other stakeholder oriented.

From a technology-oriented standpoint, authors of reviewed publications proposed adopting algorithms with promising results for mobile device applications (Alandjani 2022), implementing profile-matching systems to assist policymakers in hiring decisions and fraud prevention (Suryanto et al. 2018), and integrating contextual features to enhance the overall performance of algorithmic detection techniques (Wang et al. 2022). The subsequent discussion about study methodologies will reveal that many reviewed studies sought to devise said algorithms or assess existing algorithms, thereby potentially impacting the efficacy of the "preparation" and "pre-activity" stages of the script.

Targeting these early actions make sense; limiting them makes it more difficult for the criminal to continue the script and initiate victimisation through the four identified typologies. In a study exploring how actions from a visualisation should be prioritised for removal, Akartuna et al. (2024) found that ordering actions by the strength of their outward connections was particularly effective in disrupting a criminal ecosystem. Given the numerous activities these preparatory and pre-activity actions enable, the same logic is likely to apply here.

Government-oriented policies also focused on these preparatory and pre-activity actions. Studies suggest that policymakers may need to respond swiftly to even minor changes in employment-related crime rates, as the models predict a sharp rise in unemployment-related crimes in the early phases (Ogunmiloro et al. 2024). Scholars recommend that policymakers focus on shutting down websites associated with illegal businesses offering fake job opportunities and actively participate in the creation of hiring guidelines to combat employment fraud (Wahid 2023). Additionally, the violation of national minimum wage and working hour rules could be a valuable tool for the government to identify high-risk instances (Volodko et al. 2020).

From the perspective of other stakeholders, researchers and website hosts can work on providing training for all internet-users on spotting fraudulent opportunities and educating individuals on the ways to analyse open search data, while website hosts should promptly examine and eliminate potentially fraudulent users and

opportunities (Cole 2022). Furthermore, organisations and departments dealing with employment fraud are encouraged to conduct more extensive studies on quantifying the prevalence of recruitment fraud, identifying offender approaches and the characteristics of vulnerable individuals, and investigating the critical monitoring and detection role of job placement and posting sites (Cross and Grant-Smith 2021). Again, the focus appears to be on the preparatory and pre-activity stages of the script.

In comparison, studies rarely touched on the prevention of the "activity" stages of employment scams themselves. This is likely due to the strong psychological element that is in play at these stages, where the victim is convinced that the fraud is legitimate and is unlikely to heed to warnings to the contrary (Dove 2020). Some cases have shown that even if stakeholders (for example banks) successfully identify a victim making a transaction to a fraudster, victims may be so convinced of the legitimacy of the scam that they will override the bank's warning and proceed anyway (Solomon 2024). Both the crime script and the reviewed studies appear to agree, therefore, that prevention of victims interacting with fraudulent ads and being psychologically manipulated in the first place has better chances of success.

We now turn to study-specific insights, for example methodological approaches, gauged from our reviewed studies. This section intends to understand the methods available (and the limitations thereof) to researchers aiming to address the problems visualised in the crime script, and provides a synthesised overview of research already undertaken, where it has led thus far, and where future research may focus for maximum utility.

Study-specific insights

A brief review of study types

By closely looking at the 16 articles, we found that their aims can be roughly categorised into four types: (1) machine learning and detection research; (2) exploratory qualitative research; (3) comparative studies; and (4) country-specific studies. These will be elaborated on below.

Machine learning and detection research

Nine out of 16 papers either entirely or partially focus on either developing, improving, or testing various machine learning models on detecting fake job advertisements, and these articles are mainly serving quantitative analytical purposes.

Despite many articles aiming to study machine learning models on detecting fake advertisements within various databases, the goal of these research varies slightly. The first subgroup of research published by Ullah and Jamjoom (2023) and Kim et al. (2019) aim to develop new machine learning models to prevent future online recruitment fraud.

In contrast, the second subgroup of articles published by Vo et al. (2021) and Afzal et al. (2024) aimed to improve the effectiveness of commonly used algorithms. Specifically, Vo et al. (2021) began with testing the effectiveness of machine learning via

Table 3 An overview of the three subgroups of machine learning research on employment scams

New model development	Improving existing algorithms	Comparing algorithms/classifiers
Ullah & Jamjoom	Vo et al.	Naudé et al.
Kim et al.	Afzal et al.	Alandjani
		Wang et al.
		Srivani et al.
		Suryanto et al.

k-fold cross-validation and then proposed approaches to improve the predictability of machine learning algorithms. Additionally, Suryanto et al. (2018) focused on testing a specific algorithm, ELECTRE, on profile matching.

In comparison, the goal of the third subgroup of articles, Alandjani (2022), Wang et al. (2022), Naudé et al. (2023)[2], and Srivani et al. (2023) compared the effectiveness of commonly used algorithms or machine learning classifiers, including support vector machines, multinomial naive Bayes, decision trees, random forests, and k-nearest neighbors. Notably, Wang et al. (2022) aims to provide a detailed comparison of traditional learning and deep learning methods. Table 3 shows a summary of reviewed papers by subgroup.

Exploratory qualitative research

There are five exploratory studies examining different aspects of online recruitment fraud using qualitative research methods, including studying the tactics used by online fraudsters (Cole 2022), studying the general picture of online recruitment fraud and impact of COVID-19 on recruitment fraud (Cross and Grant-Smith 2021), learning about advanced fee fraud from the offenders' perspective (Ebot 2023), exploring the overall situation of employment fraud in Indonesia (Wahid 2023), and examining the characteristics of fake job ads in Lithuania (Volodko et al. 2020).

There are two studies, Vidros et al. (2017) and Naudé et al. (2023), that work towards more than one direction. Vidros et al. (2017) explore the characteristics of online recruitment fraud compared to other types of fraud and then evaluates the effectiveness of current common detection methods. Naudé et al. (2023) explores to what extent different categorisations of fraudulent jobs can be identified and tests the performance of four classes of features combined with various machine learning classifiers.

Country specific studies

Of the 16 studies reviewed, four targeted specific countries or regions including Nigeria, Indonesia, Cameroon, and Lithuania.

Specifically, Ogunmiloro et al. (2024) developed mathematical models to capture online job-related criminal activities in the Nigerian context. Ebot (2023) conducted a case study examining the motivations and tactics of Cameroonian scammers. Wahid

[2] Though we consider Naudé et al.'s study to be comparative, its primary research goal is the comparison of commonly used machine learning algorithms.

(2023) evaluated incidents of job scams in Indonesia. Additionally, Volodko et al. (2020) explored the characteristics of job advertisements on a Lithuanian website, which included both English and Lithuanian language listings.

Data sources

An analysis of data sources is essential to: (i) gauge the strengths and limitations of existing employment scam research; and (ii) provide an understanding of the availability of datasets for future work. Of the 16 publications, eight utilised the same dataset – namely the Employment Scam Aegean Dataset (EMSCAD). EMSCAD is a publicly accessible dataset with 17,014 real and 866 fake job ads from between 2012 and 2014. It attempts to give the scientific community a clear image of the employment scam issue and acts as a useful testing ground for scholars working in this area.

EMSCAD-contained ads can be utilised for several tasks, such as identifying phony job postings and examining important fraud signs, thereby enabling the development of classification models that can discern between fictitious and authentic job descriptions. It comprises both textual and meta-information about the jobs so that various analytical techniques can be applied to the dataset.

The utilised datasets for the remaining papers come from a variety of sources according to their specific goals or targeted regions. Ogunmiloro et al. (2024) leveraged a dataset obtained from the Nigerian Bureau of Statistics, as their research specifically focused on formulating a mathematical model to address the issue of fraud in the Nigerian context. Conversely, Cole (2022) constructed their own primary dataset, generating eight distinct resumes to examine the various types of victims targeted by job market fraudsters.

In contrast, Suryanto et al. (2018) employed a personalised MySQL database, which can be readily read, created, and modified by the researchers themselves. Ebot (2023), on the other hand, conducted a qualitative study, relying on textual data obtained through interviews.

Wahid (2023) obtained data from various sources, including legal courses and documents in Indonesia. Kim et al. (2019) used an employment agency's dataset, with the model input values comprising the job applicant's character traits, the recruitment announcement characteristics, and the counseling features, to predict the likelihood of fraudulent activity. Volodko et al. (2020) obtained their data from a specific Lithuanian website, *darbasuzsienyje.org*, which specialises in job listings for overseas employment.

Study findings

In this section, we summarise the findings of the reviewed publications as a synthesis of existing work and what it can tell us about the state of the art of employment scams and their prevention. It should be noted that, as discussed, studies were largely split into: (i) exploratory analyses into employment scams and relevant countermeasures; and (ii) identifying means of detecting scam advertisements, often through machine learning algorithmic solutions. Consequently, we split our summaries into two respective sections below.

Exploratory studies

A summary of outcomes of exploratory studies are shown in Table 4.

Detection-based studies

As previously mentioned, detection-based studies predominantly aimed to either devise new algorithmic detection solutions for fake job ads, or to compare the effectiveness of existing algorithms. Table 5 shows a summary of outcomes of these detection-based studies, including their data sources, algorithms compared, and the algorithm found to be outperforming the others.

The forthcoming discussion section will highlight the key takeaways from these studies.

Discussion

We now turn to discussing our findings and the contemporary nature of the employment scam threat. Our core aim as part of this discussion will be to suggest key gaps and future research directions that will benefit the work completed thus far in this field, based on our crime script and reviewed publications. First, we motivate the importance of such research by drawing on the serious implications that employment scams inflict on victims and financial systems, as gauged directly from reviewed publications. We then turn to discussing how this review, and its constituent publications, contribute to our overall understanding of the problem, and what the most effective next steps are in terms of future research.

Implications of employment scams and the importance of addressing them

Employment scams inflict serious harm in multiple ways, including psychological and financial distress to the victimised individual. However, on a wider basis, they serve to reduce trust in the job market, cost companies (including financial institutions and job boards) significant amounts of money in detection and mitigation efforts. They also have a significant cost burden on public finances of jurisdictions due to the additional law enforcement resources required to investigate such scams, as well as the need to fund anti-scam awareness and other such countermeasures.

The reviewed publications often noted the severe implications of not addressing employment scams in their specific contexts. From the perspective of mental well-being, being a victim of employment fraud can have a detrimental psychological impact on individuals. Financial losses, damaged trust, and the emotional toll of fraudulent activities may lead to heightened stress and anxiety, worry, and reluctance to engage in future job-seeking efforts (Afzal et al. 2024; Ogunmiloro et al. 2024).

The viewpoint of recent graduates who have been victimised by fraudulent job offers may cause job seekers to become increasingly cautious, skeptical, or even fearful of the authenticity of future job advertisements, both online and offline. This heightened vigilance may negatively impact their willingness to actively participate

Table 4 Exploratory study findings

Author	Aim	Findings and outcomes
Ogunmiloro et al. (2024)	Impact of government measures on fraudulent recruitment in Nigeria	• Higher vacancies occur with exposure of lists of ghost workers • Reduction in scam sites with strict detection and removal measures • Media-driven education on identifying and reporting job racketeers reduces risk
Cole (2022)	Exploring scammer tactics on online employment platforms	• Sensitive info scams and phishing found to be the most common tactics • Links to other financial frauds, such as cheque kiting - the time delay (float period) between when a cheque is deposited and when the funds are cleared and deducted from the issuing bank. • More than half of analysed ads had "politeness" cues to showcase legitimacy and trust • Focus target were jobseekers of low socioeconomic status
Cross and Grant-Smith (2021)	Online recruitment fraud post-Covid-19	• Need for quantification of the prevalence of employment scams • Need for clearer comparative analyses on legitimate and fake advertisements • Need for pinpointing most at-risk populations • Highlighting of the importance and responsibilities of job advertisement platforms
Suryanto et al. (2018)	Use of Elimination Et Choix Traduisant la Realite (ELECTRE), a profile-matching programme, in preventing employment fraud	• By leveraging the profile matching approach within a decision support system, managers may be able to more readily identify qualified applicants • Data generated by such tools can better help identify fraudulent advertisements • By incorporating such tools into decision support frameworks, organisations can work towards safeguarding the integrity of their hiring processes
Ebot (2023)	Criminal skill development in scammers over time	• Networking between scammers helps mutually build their skills • Peer pressure and exposure among friends engaging with illicit activity help drive further criminal engagement and normalization of scams
Wahid (2023)	Job fraud strategies and countermeasures in Indonesia	• Unemployment facilitates local recruitment fraud • Lack of awareness among workers increases scam risk • Companies facing financial hardship seek to undertake recruitment fraud to amass cheaper labour • Collaboration between government and legal institutions needed for a more effective response
Vidros et al. (2017)	Characteristics and detection challenges in online recruitment fraud	• Employment scams share traits with phishing, spam, Wikipedia vandalism, cyberbullying, trolling, and opinion fraud • Algorithms for detecting employment scams can be initially grounded in text mining techniques combined with metadata analysis • Applicant Tracking Systems (ATS) can potentially leverage datasets to train classifiers or gain deeper insights into the specifics of the problem
Volodko et al. (2020)	Indicators of labour trafficking in ads on Lithuanian job boards	• 98.4% of the examined job advertisements contained at least one indicator associated with human trafficking and majority of ads including two to four indicators • Countries that were most frequently promoted as locations for the work in the studied job boards were the Netherlands, the UK, Germany and Cyprus

Table 5 Detection-based study findings

Reference	Dataset	Dataset size	Compared algorithm(s)	Outperforming algorithm
(Alandjani 2022)	EMSCAD	17,880 real jobs ads and 866 scam ads	Support Vector Machines (SVM), Multinomial Naive Bayes, Decision Trees, K-Nearest Neighbors (K-NN), and Random Forest	K-Nearest Neighbors (K-NN)
(Vo et al. 2021)			FJD-OT	FJD-OT
(Afzal et al. 2024)			Random Forest (RF), Extremely Randomized Trees (ETC), Logistic Regression (LR), k-Nearest Neighbors (k-NN), Naive Bayes (NB), as well as a hybrid classifier.	Random Forest (RF) and Extremely Randomized Trees (ETC)
(Ullah and Jamjoom 2023)			AdaBoost (AB), Extreme Gradient Boost (XGB), Voting, and Random Forest (RF)	AdaBoost (AB)
(Srivani et al. 2023)			SVM algorithm, KNN algorithm, Random Forest algorithm, Decision Tree algorithm, Naïve bayes algorithm, and MLP algorithm	MLP algorithm—a deep neural network
(Wang et al. 2022)			Text convolutional neural network (CNN), Gate Recurrent Unit (GRU), Bidirectional Long-Short-Term Memory (BiLSTM), random forest, support vector machine (SVM), logistic regression, and Naïve Bayes	Text CNN
(Kim et al. 2019)	Data obtained from an employment agency (unspecified).	N.A.	CNN, decision trees, logistic regression, SVM, gradient boosting, random forests, hierarchical clusters - based deep neural networks (HC-DNN), Label-based Deep Neural Networks (L-DNN), Non-hierarchical Clusters-based Deep Neural Networks (NC-DNN)	HC-DNN

in the job market (Ogunmiloro et al. 2024). Moreover, the priority for job seekers to maintain the security and consistency of their personal, academic, and professional information may lead to a diminished interest in responding to new job postings (Srivani et al. 2023). Additionally, the privacy of job seekers can be compromised by recruitment fraud (Cross and Grant-Smith 2021).

From an organisational perspective, employment scams can harm the reputation of companies, such as employment websites that unknowingly advertise fraudulent positions (Afzal et al. 2024; Cross and Grant-Smith 2021). Furthermore, the task of building trust and confidence in legitimate job advertisements via social and electronic media platforms becomes increasingly challenging for legitimate organisations (Srivani et al. 2023).

Our crime script also notes that employment scams, while being a potent crime in themselves, also enable several criminal "post-activities", including identity fraud,

money mules, labour trafficking, industrial-level online scamming and dark web commerce. On a more positive note, however, it reinforces the views of reviewed publications that preventing preparatory and pre-activity actions, such as the posting of fake advertisements, may be an effective point of intervention that limits psychological harm and further criminality.

These findings all call for greater emphasis on future research on the identification and mitigation of employment scams. The next section therefore reviews the overall state of the art of employment scam research and motivates avenues for future studies.

Methodological state of the art

The review matrix in Fig. 5 shows a summary of the state of the art of employment scam research, based on the core data we have sought to chart (per Table 1). The matrix shows, through colour-coding, the areas where research has focused on more (darker green) compared to areas that have remained less explored (darker red). In addition, emboldened detection algorithms with asterisks (*) indicate algorithms that have been suggested by at least one detection-based study to outperform others. This matrix aims to showcase areas within the field of employment scams where existing research has wide coverage, compared to areas where it may be beneficial to focus on in future studies.

Some clear takeaways from this visualisation can be drawn, including the clear focus on testing established machine learning algorithms (such as random forest) on job board-based advance fee frauds within the EMSCAD database. Numerous exploratory analyses into the key indicators and causes of employment scams has also resulted in some interesting insights in a range of contexts (refer to Table 4).

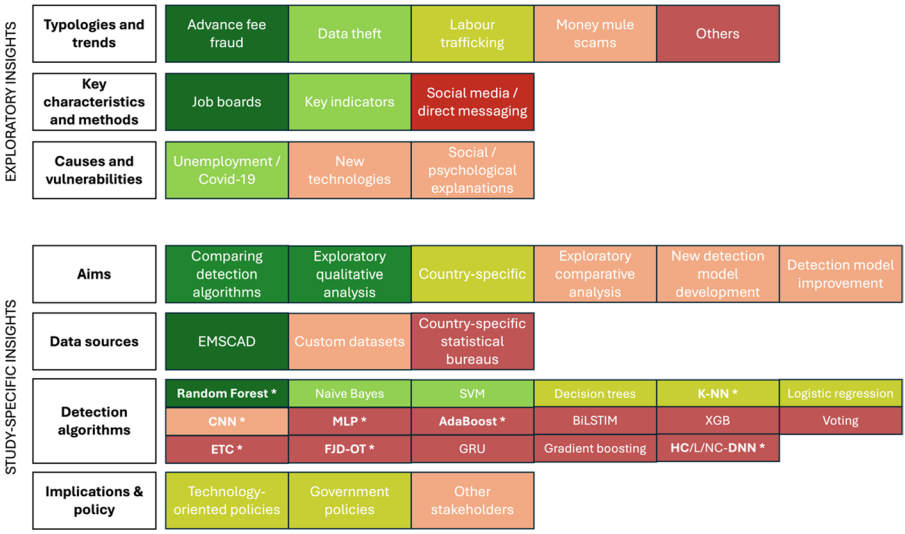


Fig. 5 Review matrix of findings, where darker green indicates more research coverage and darker red indicates less. Detection algorithms are emboldened and presented with an asterisk (*) where at least one study has found them to outperform a selection of other algorithms

There are also, however, some clear areas where further research can contribute to better understanding. For example, country-specific and labour trafficking-based employment scams have only been covered by a few publications. In many cases, algorithms that have shown promising capabilities have been tested only once, against a limited number of other algorithms. Furthermore, job scams that occur via direct messaging applications have seldom been researched, let alone mentioned, compared to fake advertisements on job boards. This likely reflects the fact that public data of message-based job scams (as opposed to job board-based ones such as EMSCAD) are not widely available.

This general state of the art reveals a range of opportunities where future research contributions would be most valuable in improving the detection and prevention of job scams. In addition, some of the reviewed publications have offered their own suggestions on avenues for future research, based specifically on their findings. Concluding our discussion, we leverage our crime script findings and research state of the art to motivate where efforts are best focused going forward.

Future research directions

Below, we consider six avenues for future research that appear, based on both the review matrix in Fig. 4 and our script, to be: (i) currently under-researched; and (ii) of particular value to developing our understanding and countermeasures against employment scams.

Country or region-specific analyses

The conceptualisation of employment scams as a country-specific phenomenon is not unheard of. Nigeria is known to have "scam schools" where such scam techniques are essentially taught to younger generations of criminals (Lazarus and Button 2024). As mentioned in the introduction, the growth of dedicated scam compounds in Southeast Asia during the Covid-19 pandemic also presents a clear regional employment scam threat to facilitate labour trafficking to countries such as Cambodia, Laos and Myanmar.

Our review matrix (Fig. 5) shows that country-specific and labour trafficking-specific employment scam research is limited. In terms of the Southeast Asian context, only one, a study on Indonesia by Wahid (2023), covered the region. No studies exist that are of direct relevance to jurisdictions that the United Nations (UNODC 2023) has since attributed to labour trafficking in the hundreds of thousands, namely Cambodia (estimated 120,000 trafficked persons), Myanmar (estimated 100,000 trafficked persons) or Laos.

Researching this phenomenon specifically is important not only due to the volume of labour trafficking it has evidently facilitated, but also because of the disruptive potential it presents to stakeholders. As our crime script shows, employment scams can be considered, essentially, the first stage of the process of establishing scam compounds that can deploy other scams on an industrialised scale. To successfully prevent the employment scam component can therefore severely restrict the capability of these compounds to reach such capabilities in the first place.

Many governments have advised against travel to the Southeast Asia region for work (Ma et al. 2025; Security Bureau, n.d.). Specific avenues for research can include appraising the effect of anti-scam awareness messaging, for example on government travel advice bulletins and job boards. Applying detection algorithms to employment ads associated specifically with the region can also assist in tailoring better detection capabilities that capture the nuances of this particular phenomenon.

Futures-oriented approaches

In the introduction, we emphasised the importance of "futures-oriented" studies in employment scams, particularly because of constant technological and global developments (such as artificial intelligence and global economic disruptions such as Covid-19) that have the potential to exacerbate such scams and make their deployment more efficient. The surge in employment scam cases in the US in 2023 was partly attributed to AI (Iacurci 2024). Specifically, as shown in our crime script, AI tools have been used to generate convincing job ads at-scale (Sherr 2024), saving criminals considerable time and manual work.

Per our review matrix, no study was futures-oriented. Furthermore, AI-generated job ads have the potential to significantly alter (or even render obsolete) the typical indicators of scam job ads on which traditional databases such as EMSCAD, as well as existing studies, are based on. This bears significant implications on the continued relevance of the sizeable amount of detection-based work that has already been done on identifying employment scams.

Futures-oriented work has a clear set of methodologies that we therefore motivate for onward research, designed primarily to identify how employment scams are likely to evolve in the future, based on anticipated global developments and trends. These include initial desk research or "horizon scanning" (of which the current study can be considered a variant), focus groups among experts, "Delphi" studies to gauge expert opinions, and broader reviews into grey literature beyond academic articles (GO Science 2024). Through such work, endeavours to develop means of detecting employment scams will be resilient to future changes and will therefore carry less risk of losing relevance as rapid technological development continues.

Further machine learning algorithmic refinement

Our crime script suggests that detecting and removing job ads at the preparatory and pre-activity stages of the employment scam process is an effective point of intervention. This reinforces the need for further refinement of existing detection algorithms and the development of new ones.

We note that a sizeable number of studies have engaged in comparing different algorithms in detecting employment scams, predominantly on the EMSCAD database. There appears, however, to be no agreement on the best performing algorithm, largely due to each study comparing different ones. This does, however, leave scope for additional comparisons between the best-performing algorithms emerging from each study, particularly because some high performing algorithms have been tested only once.

On a broader level, however, we reiterate the dangers of relying on one specific dataset with somewhat outdated entries (the EMSCAD database has collected job ads from 2012 to 2014) that do not account for more recent developments and technological advances (e.g. AI). This is because, regardless of how well an algorithm performs, it is at risk of providing little benefit or capability in identifying contemporary scam ads, which may have significantly different indicators to ones observed over a decade ago.

Dataset refinement and features identification

We note, based on our review matrix (Fig. 5), that little work has been done on employment scams disseminated through social media or direct messaging, likely due to the lack of available datasets such as EMSCAD. Our crime script suggests that this is a possible interaction method. This calls on a wider, perhaps collaborative endeavour, to collect such data on messages and engage in natural language processing analyses to identify means of detecting and stopping such advances. This could include employing advanced techniques such as Latent Semantic Analysis, the ELECTRA transformer, or word embeddings like GloVe and fastText (Naudé et al. 2023).

The importance of such work arises from two theoretical bases: first, there is no empirical evidence to suggest that the indicators of scams on job boards are equivalent to those disseminated through direct message, meaning that current detection-based studies cannot claim to be accounting for both. For message-based scams, indicators may include the use of free email solutions, offering a job without an interview, the area calling code of a WhatsApp number not matching the apparent location of the recruiter, lack of a profile picture, and so on (Seah 2023). These, at face value, bear significant differences to the scam indicators identified on job boards. Figure 6 illustrates the distinct nature of these scams.

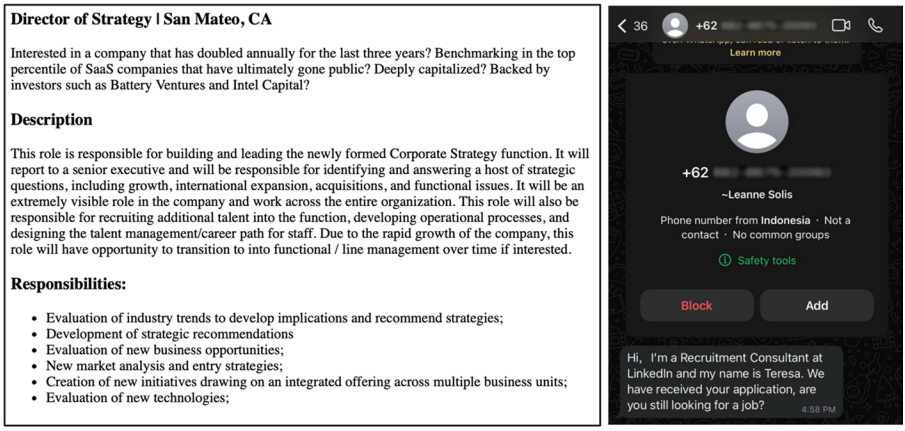


Fig. 6 A job board recruitment scam from the EMSCAD (2016) dataset (left) and a WhatsApp-based recruitment scam (right). Notice the Indonesian number and the scammer using different names for their WhatsApp name and in their message – indicators of a recruitment scam

Second, there is scope for tactical crime displacement – a phenomenon where criminals, sensing that anti-scam detection capabilities have improved (e.g. on job boards through machine learning algorithms), shift their tactics to engage in the same crime in a manner where such detection capabilities are absent (e.g. direct messaging or email) (Bowers and Johnson 2003). Thus, future research must endeavour to cover all bases, including both job board and direct messaging-based employment scams.

Crime script enhancement

As discussed, crime scripts can vary heavily in terms of detail, based on the level of available evidence (Cornish 1994). The current academic evidence base around employment scams has enabled the creation of a reasonably holistic crime script, though its individual components remain general and display clear opportunities for further granularity.

For example, the specific psychological factors invoked during advance fee fraud or data theft could be incorporated to script the actions and interactions between criminals and victims in more detail. More information regarding the operational security considerations (e.g. where criminals obtain sim cards to send fraudulent messages, how they shield their IP addresses when creating fake websites, how they create accounts to post ads on legitimate job boards) could significantly broaden the "preparation" and "pre-activity" portions of the script.

Having discussed these two categories as the most effective points of intervention, further studies into the specifics of how they work can significantly help in providing a practical understanding of how these interventions should be devised. For example, understanding the operational intricacies behind how a fake job ad is posted and the digital footprints they may exhibit can provide contextual insights for future algorithm development studies, which can improve their ability to detect fraud.

Policy analyses

Finally, we note that the number of policy implications and recommendations explicitly stated by existing studies is somewhat minimal. This is not unexpected, as many focus on developing detection algorithms with self-implied implications for employment scam prevention. However, policy-oriented studies that are designed to appraise or propose new/improved policy measures to counter contemporary employment scam risks largely remain absent.

In recent years, due to the surge in fraud across many jurisdictions (including major financial centres such as the UK (Sharp 2021), US (U.S. FTC 2024), Australia (ACCC 2024), Singapore (Sun 2024) and Hong Kong (Hong Kong Police Force 2024b), governments have enacted a range of policy responses to address the problem. For example, the UK Government (2023) declared fraud a national security risk and unveiled a national fraud strategy, with policies including partial liability for financial institutions that fail to prevent fraudulent transactions. In Hong Kong, the Police have launched an app called *Scameter+* (CyberDefender n.d.), a database of known fraudulent websites, phone numbers and bank account numbers that can be used by victims to check whether they are engaging with a scammer.

Studies focusing on appraising such interventions, verifying the extent of (and the reasons behind) their success or failure, would be of utility to promote workable policies to counter what is ultimately a global problem. Such studies would be enhanced by coming together with horizon scanning work to ensure that suggested policies and policy enhancements are resilient against technology-enabled crime displacement.

Of final note is the lack of existing studies on international co-operation – a crucial enabler (but also blockage) in countering cross-border crimes, particularly financial crimes such as scams and money laundering. The importance of cross-industry and cross-jurisdictional collaboration and data sharing in bringing financial criminals to justice is well documented (Akartuna et al. 2022; Ferwerda 2009; Thailand Business News 2024). Recalling previously mentioned regional trends such as Southeast Asian scam compounds, studies focusing on enhancing such cross-border collaboration would be of increasingly high relevance.

Limitations

This review has a limitation. First, it has limited its scope to academic articles in English, while acknowledging that grey literature and other relevant sources (such as news articles cited throughout the introduction and discussion) also have considerable insights to contribute. On this basis, it is acknowledged that some research gaps identified throughout our discussion may have been addressed through mediums beyond academia.

We justify our focus on academia on the basis that our research question specifically intends to understand the nature and potential of academic contributions in countering the surge in employment scams. Its limitations (such as publication delays) notwithstanding, academic work provides a unique contribution of qualitative and quantitative insights, which is crucial for assisting with exploratory work, horizon scanning, devising detection algorithms and policy recommendations.

Conclusion

Employment scams remain a significant problem across numerous jurisdictions. Not only do they cause significant psychological and financial harm on victims, they also reduce trust in employment drives, job boards and the financial system as a whole. In addition, employment scams have become a foundational enabler of more serious criminality, as demonstrated by large-scale labour trafficking enabling industrialised scam compounds in Southeast Asia.

On this basis, it is crucial to understand the state of the art and consolidate ideal ways forward to ensure that future research remains of maximum benefit to stakeholders with preventative responsibilities. To this end, we have synthesised existing research into a visual accessible crime script to holistically understand the various employment scam typologies and methods involved. We have also found that existing literature on the topic covers a range of objectives, ranging from exploratory analyses to comparing detection algorithms. We have noted the contributions but

also limitations of such approaches, including the reliance of a single and outdated dataset, the general lack of geographical focus and the need for more futures-oriented and policy-relevant research.

As a result, we propose six consolidated avenues for future directions, namely: (1) more country/region-specific analyses to account for recent trends in Southeast Asia in particular; (2) futures-oriented and horizon scanning approaches to account for how employment scams may develop in the future; (3) further refinement of detection algorithms; (4) further work on refining datasets and features identification for message/email-based scams that are currently under-researched; (5) enhancing our crime script with greater granularity; and (6) more policy-oriented research and appraisals. We motivate these strands as powerful research directions that can help maximise academic contributions to detecting and mitigating the surging employment scams of today and tomorrow.

Author contributions All authors whose names appear on the submission made substantial contributions to the conception or design of the work; the acquisition, analysis, and interpretation of data; drafted the work or revised it critically for important intellectual content; approved the version to be published; and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability Much of the collected data throughout the review is readily available in Tables 3, 4 and 5 of this article. If the reader wishes for the data in another format, they may contact the authors.

Declarations

Ethics, Consent to Participate, and Consent to Publish Not applicable.

Competing interests The authors declare no competing interests.

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